

PINK REPORT

This table summarizes the available evidence on **gabapentin for overall survival (OS) in glioblastoma**, organized from highest to lowest level of rigor. Current standards-of-care guidelines **do not recommend gabapentin for OS benefit**, while **observational human data** suggest an association with improved survival that remains **investigational** (NCCN; Weller; Bernstock; Mehari).

Recommendation (guided by the evidence): Use **gabapentin only for standard symptom-based indications** (e.g., neuropathic pain, seizures if needed), **not** with the primary goal of improving OS at this time, because the OS signal is supported only by **retrospective/observational data** and **no randomized efficacy results** are yet available (NCCN; Bernstock; Mehari; Mastall).

References supporting summary ordered by strength of evidence:

Rank	Level of Evidence	Study Title	Authors	Journal (Year)	Notes
 1	Guideline	<i>Central Nervous System Cancers</i>	National Comprehensive Cancer Network (NCCN)	NCCN Guidelines (Updated 2025)	Does not include gabapentin as an anti-tumor/OS-improving therapy for glioblastoma; lists it as an option among non-enzyme-inducing AEDs (primarily for seizure-related indications). (NCCN)
 2	Clinical Trial Meta-analysis				(No eligible meta-analysis identified in provided references.)
  3	Phase 3 (P3) Trial				(No P3 trial evidence identified in provided references.)
 4	Phase 2 (P2) Trial				(No completed P2 efficacy trial evidence identified in provided references.)
 5	Phase 1 (P1) Trial	<i>A Phase Ib/II Randomized, Open-Label Drug Repurposing Trial of</i>	Mastall et al.	BMC Cancer (2024)	Trial protocol (Phase Ib/II) evaluating glutamate-signaling inhibitors (including gabapentin in the

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		<i>Glutamate Signaling Inhibitors... (GLUGLIO) Trial Protocol</i>			rationale/repurposing context); results pending , so no prospective OS benefit established. <i>(Mastall)</i>
6	Large Cohort Study	<i>Gabapentinoids Confer Survival Benefit in Human Glioblastoma</i>	Bernstock et al.	Nature Communications (2025)	Observational study (reported n≈1,072) associating gabapentin exposure with improved survival and reduced serum TSP-1; association ≠ causation ; not yet practice-changing. <i>(Bernstock)</i>
7	Cohort Study	<i>Can targeting TSP-1 with gabapentin enhance survival in glioblastoma? A 20-year retrospective cohort study</i>	Mehari et al.	J Clin Oncol (ASCO abstract) (2024)	Retrospective association: OS difference reported; median daily dose reported around 600 mg/day (IQR 300–900) ; abstract-level evidence; confounding possible; limited generalizability to frail ECOG 3 patients. <i>(Mehari)</i>
8	Animal Model				<i>(No animal-model evidence identified in provided references.)</i>